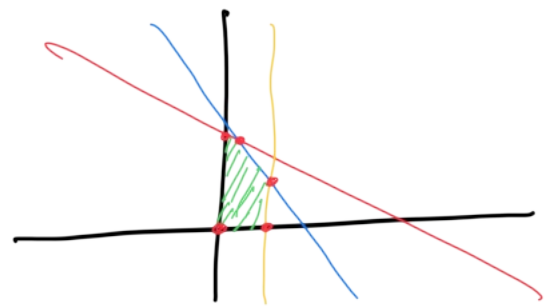


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- Graphical Method
 - Used for solving small-scale optimization problems (mostly 2 variables).
 - Feasible region is plotted based on constraints.
 - Objective function (like $z = ax + by$) is moved parallelly to find the optimal point.
 - Optimum lies at a corner point (vertex) of the feasible region.

- Vector and Tuple, Vector Space
 - A tuple is an ordered list (e.g., (x, y, z)), a vector is a geometric object with magnitude and direction.
 - Vector space: A set of vectors closed under vector addition and scalar multiplication.
 - Must satisfy properties like commutativity, associativity, existence of zero vector, etc.

- Convex Set, Convex Combination
 - Convex set: A set is convex if for any two points x, y , the line segment $(1 - \lambda)x + \lambda y$ lies in the set, for $0 \leq \lambda \leq 1$.
 - Convex combination: A linear combination $\sum \lambda_i x_i$, where $\lambda_i \geq 0$ and $\sum \lambda_i = 1$.

- Convex Function ($\cup$ shaped), Concave Function ($\cap$ shaped)
 - Convex function: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$
 - Looks $\cup$ -shaped; local min = global min.
 - Concave function: $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$
 - Looks $\cap$ -shaped; local max = global max.

Question

$$\begin{array}{ll} \textcircled{1} & \text{Min } 3x + 2y \\ \text{s.t.} & x + y \leq 4, x \geq 0, y \geq 0 \end{array}$$

$$\begin{array}{ll} \textcircled{2} & \text{Min } x + 2y \\ \text{s.t.} & x + y \geq 2 \\ & 2x + y \geq 4, x \geq 0, y \geq 0 \end{array}$$

$$\begin{array}{ll} \textcircled{3} & \text{Min } 5x + 4y \\ \text{s.t.} & x + 2y \leq 20 \\ & 3x + y \leq 30 \\ & x \geq 0, y \geq 0 \end{array}$$

$$\begin{array}{ll} \textcircled{4} & 2x + y \\ \text{s.t.} & x + y \geq 3 \\ & x + 2y \geq 4 \\ & x \geq 0 \\ & y \geq 0 \end{array}$$

- Hessian Matrix

- Square matrix of second-order partial derivatives:

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

- Used to analyze curvature of multivariate functions.

Give

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

- Determinant Calculations

- Determinant tells us:

- Area/volume transformation.
- Whether a matrix is invertible ($\det \neq 0$).
- Useful in checking definiteness of matrices (used in optimization).

$$D_1 = [a_{11}]$$

$$D_2 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$D_2 = a_{11} \times a_{22} - a_{12} \times a_{21}$$

$$D_3 = 0 = a_{11} \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} + a_{13} \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

- Symmetric Matrix

- A matrix is symmetric if $A = A^T$.
- All eigenvalues of symmetric matrices are real.
- Hessians of twice differentiable scalar functions are symmetric.

Rows & Column same

- Positive/Negative Definite Hessian (Local Minima/Maxima)

- Positive definite: All eigenvalues $> 0 \rightarrow$ local minimum.
- Negative definite: All eigenvalues $< 0 \rightarrow$ local maximum.
- Indefinite: Mixed eigenvalues $\rightarrow$ saddle point.

$> > > 0 \dots$

$> < > < \dots$

- Convexity/Concavity via Hessian

- Convexity: If Hessian is positive semi-definite, function is convex.
- Concavity: If Hessian is negative semi-definite, function is concave.
- Strict convexity/concavity when Hessian is positive/negative definite.

- Global Minima/Maxima
 - In convex optimization:
 - Any local min is a global min if the function is convex.
 - For non-convex functions, multiple local minima or saddle points may exist.
 - Sufficient condition: Convex objective + convex feasible region $\Rightarrow$ global optima.
-

- Orthogonal Vectors
 - Two vectors $\vec{u}, \vec{v}$ are orthogonal if $\vec{u} \cdot \vec{v} = 0$.
 - Basis vectors in Euclidean space are often orthogonal (and orthonormal).
 - Orthogonality implies linear independence (but converse not true).

Let me know if you want quick examples or visual interpretations of any part!

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Method ①

First & Second Order Conditions for Optimality (Unconstrained Optimization)

First-Order Necessary Condition (FONC)

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, if x^* is a local minimum (or maximum) and f is differentiable at x^* , then:

- Gradient condition:

Step ①

$$\nabla f(x^*) = 0$$

$$\frac{df}{dx} = 0$$

$$\frac{df}{dy} = 0$$

This means the directional derivative of f in every direction is zero; geometrically, the surface is flat (no slope) at x^* .

Step ②

$$\frac{d^2f}{dx^2}, \frac{d^2f}{dy^2}, \frac{d^2f}{dx, dy}$$

Second-Order Conditions

These involve the Hessian matrix, $\nabla^2 f(x)$, i.e., the matrix of second partial derivatives.

Second-Order Necessary Condition (SONC)

If x^* is a local minimum and f is twice continuously differentiable:

- $\nabla f(x^*) = 0$
- $\nabla^2 f(x^*) \succeq 0$ (Hessian is positive semi-definite)

This implies the surface is flat at x^* , and curves upwards in all directions, or at least not downwards.

Second-Order Sufficient Condition (SOSC)

If f is twice continuously differentiable, and:

- $\nabla f(x^*) = 0$
- $\nabla^2 f(x^*) \succ 0$ (Hessian is positive definite)

Then, x^* is a strict local minimum.

Summary Table

ent $\nabla f(x^*)$	Hessian $\nabla^2 f(x^*)$	Implication
0	—	Stationary point
0	Positive semi-definite	Necessary for local minimum
0	Positive definite	Sufficient for local min

Geometric Interpretation

- At a local minimum, the function "curves up" — like a bowl.
- FONC gives candidate points (stationary), but to confirm it's truly a minimum, we need second-order info about curvature.
- A positive definite Hessian implies the surface is "strictly convex" at that point — a true valley.

Extra Notes *(Ignore)*

- For convex functions, FONC is also sufficient for global minimum.
- Negative definite Hessian → local maximum.
- Indefinite Hessian → saddle point (neither min nor max).

Question

① $f(x,y) = x^2 + y^2 + 2xy$

② $f(x,y) = x^3 - 3xy^2$

③ $f(x,y) = x^4 + y^4 - 4xy + 1$

④ $f(x,y) = x^2 - y^2$

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Method
②

Line Search Method – Key Concepts & Classical Techniques

Unimodal Function *(theory only)*

- A function is unimodal on an interval if it has a single minimum (or maximum) in that interval.
- Formally, for minimization:
 - $\exists$ a point $x^* \in [a, b]$ such that:
 - $f(x)$ is decreasing on $[a, x^*]$
 - $f(x)$ is increasing on $[x^*, b]$
- Assumption of unimodality is essential for most line search methods to guarantee convergence to a global minimum in 1D.

Line Search – General Framework

- Goal: Minimize a function $f(x)$ over a 1D interval $[a, b]$
- Process:
 1. Bracketing the minimum: Start with an interval $[a, b]$ such that minimum lies within.
 2. Reduction strategy: Iteratively shrink the interval using function evaluations until the length $|b - a| < \epsilon$ (tolerance).

Method

Golden Section Method

- Based on the golden ratio: $\phi = \frac{\sqrt{5}-1}{2} \approx 0.618$
- Uses fixed ratio to maintain consistent reduction pattern.
- Steps:
 - Initialize: $x_1 = a + (1 - \phi)(b - a)$, $x_2 = a + \phi(b - a)$
 - Evaluate $f(x_1), f(x_2)$
 - Discard the worse half (where f is larger) *in Min*
 - Repeat with new interval
- Number of iterations to reach an interval of size ϵ :

do untill diff $< \epsilon$ (error)

$$n = \left\lceil \frac{\ln(\epsilon/L)}{\ln(\phi)} \right\rceil, \quad L = \text{initial interval length}$$

- Pros:
 - No need to know number of iterations in advance.
 - Only one new function evaluation per iteration (reuses old values).

Method

Fibonacci Search Method

same just a lightly change

- Similar in philosophy to Golden Section but uses Fibonacci numbers for more precise interval reduction.
- Requires pre-determined number of iterations n
- Ratio of interval reduction at each step is guided by:

$$x_1 = a + \frac{F_{n-2}}{F_n} (b - a), \quad x_2 = a + \frac{F_{n-1}}{F_n} (b - a)$$

where F_n is the n -th Fibonacci number.

- Function is evaluated at these points and the interval is reduced accordingly.
- The number of iterations n is chosen such that:

$$F_n > \frac{b - a}{\epsilon}$$

- Pros:
 - Slightly more efficient than golden section for known precision
 - Always terminates in exactly n steps
- Cons:
 - Need to fix number of steps before the algorithm begins

Summary Table

Evaluation Strategy	Needs Iteration Count?	Efficiency
Fixed reduction via ϕ	No	Moderate
Optimized via Fibonacci	Yes	High

When to Use What?

- Golden Section: Preferred when you want simplicity and don't know in advance how many steps to use.
- Fibonacci: Ideal when you must minimize function evaluations and the error tolerance is known in advance.

These methods are widely used in univariate optimization, especially in line search steps of more complex multivariate algorithms like steepest descent, Newton's method, and conjugate gradient methods.

Question Golden Method

① $\min f(x) = (x-2)^2$ in $[1, 3]$
 $\text{error} \leq 0.42$

② $\min f(x) = -x^2 + 4x$ in $[0, 5]$
 $\text{error} < 0.1$

③ $\min f(x) = x^4 - 14x^2 + 24x$ in $[0, 4]$
 $\varepsilon = 0.2$

Fibonacci Method

① $f(x) = (x-2)^2$ in $[1, 3]$
 $\varepsilon \leq 0.42$

② $f(x) = 3x^2 - e^x$ in $[0, 1]$
 $\varepsilon \leq 0.05$

③ $f(x) = x^4 - 14x^2 + 24x$ in $[0, 4]$, $\varepsilon \leq 0.2$

④ $f(x) = x^2 + 1$ in $[-2, 2]$
 $\varepsilon < 0.1$

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Method

(theory only)

- **Descent Method**

- General iterative method to find local minima.
- Update rule:

$$x_{k+1} = x_k + \alpha_k d_k$$

where d_k is a descent direction such that $\nabla f(x_k) \cdot d_k < 0$.

- Ensures that $f(x_{k+1}) < f(x_k)$ under a proper step size α_k .

understanding

- **Taylor Series Expansion**

- Approximation of a function around point x using its derivatives.
- First-order:

$$f(x + d) \approx f(x) + \nabla f(x)^T d$$

- Second-order:

$$f(x + d) \approx f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x) d$$

- Basis for Newton and Quasi-Newton methods.

Method

- Newton's Method

- Uses second-order approximation to find stationary points.
- Update rule:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

- Requires Hessian $\nabla^2 f(x_k)$ to be invertible and positive definite.
- Quadratic convergence near solution.

Q

- Single Variable Newton Method

- Special case for scalar function $f : \mathbb{R} \rightarrow \mathbb{R}$.
- Iteration:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

- Rapid convergence if starting point is near the root.

①

• Multivariable Extension

- Generalization of Newton's method to vector-valued functions.
- Uses gradient $\nabla f(x)$ and Hessian $\nabla^2 f(x)$ in update rule.
- Sensitive to initial point and may diverge if Hessian is indefinite.

②

• Steepest Descent Method

- Chooses direction of steepest decrease $d_k = -\nabla f(x_k)$.

- Update:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

- Slower than Newton's method, especially in ill-conditioned problems.
- Converges linearly.

$$D_u f(x_0, y_0) = \nabla f(x_0, y_0) \cdot u$$

$u = \frac{v}{\|v\|}$, v is given vector

• Directional Derivatives

- Measures rate of change of function in direction d :

$$D_d f(x) = \nabla f(x)^T d$$

- Positive: function increases in direction d .
Negative: function decreases in direction d .
- Descent direction: $\nabla f(x)^T d < 0$.

③

• Quasi-Newton Method

- Approximates inverse Hessian to avoid computing it directly.
- DFP update (Davidon-Fletcher-Powell):
 - Start with $B_0 = I$ (identity matrix).
 - Iterate with:

$$d_k = -B_k \nabla f(x_k)$$

$$x_{k+1} = x_k + \alpha_k d_k$$

$$B_{k+1} = B_k + \frac{ss^T}{s^T y} - \frac{B_k y y^T B_k}{y^T B_k y}$$

where $s = x_{k+1} - x_k, y = \nabla f(x_{k+1}) - \nabla f(x_k)$.

- Superlinear convergence. Efficient for large-scale problems.

Newton 1-variable

$$f(x) = x^2 + 4x + 4, \quad x_0 = 2$$

Solve

$$f'(x) = 2x + 4$$

$$f''(x) = 2$$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)} = 2 - \frac{2x_0 + 4}{2}$$

$$= 2 - \frac{2 \times 2 + 4}{2} = 2 - 4 = -2$$

$$x_1 = -2$$

$$x_2 = x_1 - \frac{2x_1 + 4}{2} = -2 - \frac{2 \times (-2) + 4}{2}$$

$$= -2 - 0 = -2$$

$$x_3 = x_2 \quad \text{stop}$$

$x = -2$ is minima ds

$$\begin{aligned} f(x) &= x^2 + 4x + 4 \\ &= (-2)^2 + 4(-2) + 4 \\ &= 4 - 8 + 4 = 0 \quad \text{h} \end{aligned}$$

Newton Multi

$$q \quad f(x, y) = x^2 + y^2 + xy + x$$

$$(x_0, y_0) = (1, 1)$$

Solve

$$x_1 = x_0 - H^{-1} \nabla f$$

$$\frac{df}{dx} = 2x + y + 1, \quad \frac{df}{dy} = 2y + x$$

$$\nabla f = \begin{bmatrix} 2x + y + 1 \\ 2y + x \end{bmatrix} \quad \text{at } (1, 1)$$

$$\nabla f(1, 1) = \begin{bmatrix} 2 \times 1 + 1 + 1 \\ 2 \times 1 + 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad D = 2 \times 2 - 1 \times 1 = 3$$

Inverse of Matrix

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad D(M) = ad - bc$$

$$M^{-1} = \frac{1}{D} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$H^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2(4) + (-1)(3) \\ (-1)(4) + 2 \times 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - 5/3 \\ 1 - 2/3 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 1/3 \end{bmatrix}$$

$$x_1 = (-2/3, 1/3)$$

Steepest Descent Method

$$x_1 = x_0 - \alpha_0 \nabla f(x_0)$$

Q1) $x^2 + y^2$, $(2, 2)$

Q2) $3x^2 + y^2 - 2xy + 4x$, $(1, -1)$

Q3) $x^2 + y^2 + z^2 + xy + yz$, $(1, 1, 1)$

Q4) $e^x + y^2$, $(0, 1)$

Solve Q1

$$(x_0, y_0) = (2, 2)$$

$$\frac{df}{dx} = 2x, \quad \frac{df}{dy} = 2y$$

$$\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix}, \quad \nabla f(2, 2) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Descent direction

$$d_0 = -\nabla f = \begin{bmatrix} -4 \\ -4 \end{bmatrix}$$

$$x_1 = x_0 + \alpha d_0$$

$$= (2, 2) + \alpha(-4, -4)$$

$$= (2 - 4\alpha, 2 - 4\alpha)$$

$$f(x_1) = f(\alpha)$$

$$f(\alpha) = (2 - 4\alpha)^2 + (2 - 4\alpha)^2$$

$$= 2(2 - 4\alpha)^2$$

$$\frac{df(\alpha)}{d\alpha} = 4(2 - 4\alpha)(-4) = 0$$

$$2 - 4\alpha = 0 \Rightarrow \alpha = 2/4 = 0.5$$

$$x = x_0 - \alpha_0 \nabla f(x_0)$$

$$x = 2 - 0.5(-4) = 0$$

$$y = 2 - 0.5(-4) = 0$$

$$(x_1, y_1) = (0, 0)$$

Quasi Newton / DFP Method)

Modify version of steepest decent

$$x_1 = x_0 - \alpha B_0 \nabla F(x_0)$$

Q ① $x^2 + y^2 + 2xy$ (1,1) , $B_0 = I$
② $2x^2 + xy + y^2$ (1,0) "
③ $4x^2 + 9y^2$ (1,1) "

Solve ①

$$\nabla F(x, y) = \begin{bmatrix} 2x+2y \\ 2x+2y \end{bmatrix}$$

$$\nabla F(1,1) = \begin{bmatrix} 2 \times 1 + 2 \times 1 \\ 2 \times 1 + 2 \times 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$d_0 = -B_0 \nabla f_0 = - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ -4 \end{bmatrix}$$

$$\begin{aligned} x(\alpha) &= x_0 + \alpha (-B_0 \nabla f_0) \\ &= (1,1) + \alpha (-4, -4) \\ &= (1-4\alpha, 1-4\alpha) \end{aligned}$$

$$\begin{aligned} f(\alpha) &= (1-4\alpha)^2 + (1-4\alpha)^2 + 2(1-4\alpha)^2 \\ &= 4(1-4\alpha)^2 \end{aligned}$$

$$\frac{df(\alpha)}{d\alpha} = 8(1-4\alpha)(-4) = 0$$
$$1-4\alpha = 0 \Rightarrow \alpha = 1/4 = 0.25$$

$$x = 1 - 4 \times 0.25 = 0$$

$$y = 1 - 4 \times 0.25 = 0$$

$$(x_1, y_1) = (0, 0)$$

$$y = \nabla f_1 - \nabla f_0$$

$$s = x_1 - x_0$$

$$B_1 = B_0 + \frac{ss^T}{s^T y} - \frac{B_0 y y^T B_0}{y^T B_0 y}$$

Conjugate Gradient Method & Conjugate Directions

Conjugate Gradient Method (CGM)

The Conjugate Gradient Method is an iterative optimization technique primarily used to solve unconstrained minimization problems of quadratic functions, especially when the dimensionality is large and the matrix involved is sparse and symmetric positive definite.

Problem Setup

Minimize a quadratic function:

$$f(x) = \frac{1}{2} x^T A x - b^T x$$

where A is symmetric and positive definite.

Algorithm Overview

Let x_0 be an initial guess, $r_0 = b - Ax_0$ (initial residual), and $d_0 = r_0$ (initial direction).

Iterative steps for $k = 0, 1, 2, \dots$:

1. Step size:

$$\alpha_k = \frac{r_k^T r_k}{d_k^T A d_k}$$

2. Update estimate:

$$x_{k+1} = x_k + \alpha_k d_k$$

3. Update residual:

$$r_{k+1} = r_k - \alpha_k A d_k$$

4. Direction update coefficient:

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}$$

5. Update direction:

$$d_{k+1} = r_{k+1} + \beta_k d_k$$

6. Repeat until $\|r_k\|$ is sufficiently small.

Key Properties

- It converges to the exact solution in at most n steps for n -dimensional problems (in exact arithmetic).

- Exploits the structure of quadratic problems; more efficient than steepest descent, especially for ill-conditioned problems.
 - No need to store or invert matrix A ; only matrix-vector multiplications are used.
-

Conjugate Directions

Two non-zero vectors d_i and d_j are said to be conjugate (with respect to matrix A) if:

$$d_i^T A d_j = 0$$

That is, they are A-orthogonal.

Generalization

A set of directions $\{d_0, d_1, \dots, d_{n-1}\}$ is said to be conjugate if:

$$d_i^T A d_j = 0 \quad \forall i \neq j$$

Use of Conjugate Directions

- If you move successively along conjugate directions using exact line search, then you minimize the quadratic function in at most n steps.
 - CGM is designed to construct these conjugate directions on the fly, avoiding the need to compute them in advance.
-

Comparison to Steepest Descent

- Steepest Descent chooses direction based on the negative gradient. It may "zig-zag" and converge slowly.
 - CGM uses orthogonal residuals and A-conjugate directions to accelerate convergence, reducing redundant movement.
-

Application Tip

- Best suited for solving systems like $Ax = b$ where A is large, sparse, symmetric, and positive-definite.
- Useful in machine learning, physics simulations, and large-scale optimization.

~~Q~~ ① $f(x, y) = 2x^2 + 2xy + y^2 - 6x - 6y$

② $Ax = b$
 $A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

③ $f(x) = \frac{1}{2} x^T \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} x - \frac{1}{2} \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T x$

Least Square Approximations – Core Concepts & Key Insights

◆ Purpose

Least Squares Approximation is used to find the "best fit" solution to overdetermined systems—i.e., when the number of equations exceeds the number of unknowns. It minimizes the sum of the squares of the residuals (differences between observed and estimated values).

◆ Problem Setup

Given a system of equations:

$$Ax = b$$

Where:

- $A \in \mathbb{R}^{m \times n}$ (with $m > n$: more equations than variables)
- $b \in \mathbb{R}^m$ is the observation vector
- The system may not have an exact solution

Objective:

$$\min_x \|Ax - b\|^2$$

◆ Solution Approach

The optimal solution x^* satisfies the normal equations:

$$A^T Ax = A^T b$$

- $A^T A$: square and symmetric matrix
- $A^T b$: projection of b into the column space of A

This converts the overdetermined system into a solvable $n \times n$ system.

◆ Geometric Interpretation

- Ax lies in the column space of A
- Ax is the orthogonal projection of b onto the column space of A
- Residual $r = b - Ax$ is orthogonal to column space of A

$$A^T(b - Ax) = 0 \quad (\text{orthogonality condition})$$

◆ Key Properties

- Solution is unique if $A^T A$ is invertible (i.e., A has full column rank)
 - Least squares estimate minimizes error in Euclidean norm
-

(Theory)

◆ Computational Methods

- Direct: Solve $A^T Ax = A^T b$ via LU or Cholesky decomposition
- QR Decomposition: Preferred numerically:

$$A = QR \Rightarrow Rx = Q^T b$$

- SVD (Singular Value Decomposition): Most stable for ill-conditioned systems
-

◆ Applications

- Data fitting (linear regression)
 - Signal processing
 - Control systems
 - Machine learning (e.g., linear models, PCA)
-

◆ Extension

Weighted Least Squares (WLS):

$$\min_x \|W(Ax - b)\|^2$$

Where W is a diagonal matrix of weights to control the influence of each residual.

Regularized Least Squares (Ridge Regression):

$$\min_x \|Ax - b\|^2 + \lambda \|x\|^2$$

Used to handle multicollinearity or overfitting.

◆ Error Metric

Residual error:

$$\text{Error} = \|Ax - b\|^2 = (Ax - b)^T (Ax - b)$$

Measures the squared distance between actual data and predicted data.

◆ Visual Insight

Think of finding the closest point Ax on a flat plane (column space of A) to a point b in higher-dimensional space. The least squares solution drops a perpendicular from b to that plane.

Sourced from: Lecture_8_Least_square_approximations.pdf.

Q1) Fit line $y = mx + c$ from points $(1, 2)$ $(2, 3)$ $(3, 5)$ $(4, 4)$

Q2) $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Solve 2

$$\min_x \|Ax - b\|^2$$

calculate

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 1 & 6 \\ 6 & 14 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Solve

$$A^T A x = A^T b$$

$$\begin{bmatrix} 3 & 6 \\ 8 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$2(3x_1 + 6x_2 = 5)$$

$$6x_1 + 14x_2 = 11$$

$$\begin{array}{r} 6x_1 + 12x_2 = 10 \\ - \quad 6x_1 + 14x_2 = 11 \\ \hline -2x_2 = -1 \end{array}$$

$$x_2 = 0.5$$

$$3x_1 + 6 \times 0.5 = 5 \Rightarrow 3x_1 = 2$$

$$x_1 = 2/3 = 0.667$$

(0.667, 2) is solution

Gist on: Simplex & Big M Method

Conversion to Equalities

- Inequality constraints ($\leq$, $\geq$) must be converted to equations for the simplex tableau.
 - Do this by introducing slack, surplus, and artificial variables:
 - $\leq$ → Add slack variable (e.g., $+s_1$)
 - $\geq$ → Subtract surplus and add artificial variable (e.g., $-s_1 + a_1$)
 - $=$ → Add artificial variable only (e.g., $+a_1$)
-

Simplex Method - Core Idea

- Iterative tabular process to move from one basic feasible solution (BFS) to another, improving the objective function at each step.
 - Basic variables are the current solution set; non-basic variables are zero.
-

$c_j - z_j$ Condition

- c_j : Coefficient of variable in the objective function.
 - z_j : Net evaluation row from basic variables.
 - Compute $(c_j - z_j)$:
 - If maximization: pick variable with most positive $(c_j - z_j)$
 - If minimization: pick variable with most negative $(c_j - z_j)$
 - Optimality condition met when all $(c_j - z_j) \leq 0$ (for max)
-

Pivot Element Selection Condition

- Select entering variable: Column with largest positive $(c_j - z_j)$.
 - Select leaving variable: Apply the minimum ratio test:
 - Compute $\frac{\text{RHS}}{\text{Pivot Column Positive Element}}$
-

- Row with minimum ratio $\rightarrow$ pivot row
 - The element at the intersection of pivot row and entering column = pivot element
-

Entry Edit Condition (Tableau Update)

- New Pivot Row = Pivot row $\div$ Pivot element
 - For all other rows:
 - New Row = Old Row – (corresponding pivot column coefficient) $\times$ (New Pivot Row)
 - Repeat until all $(c_j - z_j)$ conditions are met.
-

Big M Method (When Artificial Variables Are Used)

- Used when constraints require artificial variables (e.g., $\geq$ or $=$)
 - Modify objective function:
 - Maximize: $Z = Cx - MA$
 - Minimize: $Z = Cx + MA$
 - M is a very large constant
 - Artificial variables are penalized heavily to drive them out of the basis.
-

This method ensures feasibility from the start (via artificial variables) while steering toward feasibility and optimality by penalizing artificial participation in the objective.

Q1)

$$\max Z = 3x + 5y$$

$$\text{s.t. } 2x + y \leq 18$$

$$2x + 3y \leq 42$$

$$3x + y \leq 24$$

$$x, y \geq 0$$

Q2)

$$\max Z = 5x + 3y$$

$$\text{s.t. } 2x + y = 8$$

$$x + y \geq 6$$

$$x, y \geq 0$$

Q3)

$$\max Z = 4x_1 + 2x_2$$

$$\text{s.t.}$$

$$x_1 + x_2 \leq 3$$

$$2x_1 + x_2 \geq 5$$

$$x_1 + 3x_2 = 6$$

$$x_1, x_2 \geq 0$$

Lagrange Multipliers & KKT Conditions

1. Lagrange Multipliers: Method for Constrained Optimization

Used for solving problems of the form:

- Minimize/Maximize: $f(x)$
- Subject to: $g_i(x) = 0$, where $i = 1, 2, \dots, m$

Core Idea

Convert a constrained problem into an unconstrained one by incorporating the constraint(s) into the objective using multipliers λ_i :



$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

Necessary Conditions

- Stationarity: $\nabla_x \mathcal{L}(x, \lambda) = 0$
- Feasibility: $g_i(x) = 0$

Interpretation

- At the optimal point, the gradients of the objective and constraints are linearly dependent.
- The Lagrange multipliers λ_i represent the rate at which the optimal value of the objective function increases with a unit increase in $g_i(x)$.

2. Karush-Kuhn-Tucker (KKT) Conditions

Generalization of Lagrange Multipliers to nonlinear optimization with inequality constraints.

Problem Form

- Minimize $f(x)$
- Subject to:
 - $g_i(x) \leq 0, i = 1, \dots, m$ (inequality)
 - $h_j(x) = 0, j = 1, \dots, p$ (equality)

Lagrangian

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x)$$



KKT Conditions

1. Stationarity:

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) + \sum_{j=1}^p \mu_j \nabla h_j(x^*) = 0$$

2. Primal Feasibility:

$$g_i(x^*) \leq 0, \quad h_j(x^*) = 0$$

3. Dual Feasibility:

$$\lambda_i \geq 0$$

4. Complementary Slackness:

$$\lambda_i g_i(x^*) = 0 \quad \text{for all } i$$

3. Summary Table: Comparison

Handles	Conditions
constraints only	$\nabla f + \sum \lambda \nabla g = 0, g(x) = 0$
ality + Equality	Adds dual feasibility & complementary slackness
KKT-specific	Enforces that either constraint is tight or multiplier is zero

4. Geometric Insight

- At optimum, the contour of the objective is tangent to the feasible set boundary.
- Lagrange multipliers act as weights ensuring the trade-off between objective and constraints.

5. Applications

- Support Vector Machines (SVMs)
- Portfolio Optimization

- Resource Allocation
- Economics (utility maximization, cost minimization)

6. Notes on Sufficiency

- If f, g_i, h_j are convex, and KKT conditions are satisfied, then x^* is a global minimum.
- For non-convex problems, KKT gives necessary but not sufficient conditions.

Lagrange Multiplier

Q1) $\max Z = xy$
 $s.t. \quad x+y = 10$

2) $x^2 + y^2, \quad s.t. \quad x+y = 4$

3) $xy^2, \quad s.t. \quad x+y+z = 12$

KKT

Q1) $\min x^2 + y^2, \quad x+y-1 \leq 0$

2) $\min (x-2)^2 + (y-3)^2, \quad x+y \leq 4$

3) $\max xy$
 $s.t. \quad x+y = 1, \quad x \geq 0, y \geq 0$

ODS by Manvendra

Penalty and Barrier Methods

These are techniques used in constrained optimization to transform constrained problems into unconstrained ones so that standard optimization algorithms (like gradient descent, Newton's method) can be applied.

Penalty Methods

- Core Idea: Penalize constraint violations by adding a penalty term to the objective function. The optimization is carried out over an unconstrained space, and constraint satisfaction is *enforced indirectly* through heavy penalties.
- Formulation: For problem:

$$\min f(x) \quad \text{subject to } c_i(x) = 0, i = 1, \dots, m$$

The penalized problem becomes:

$$\min \Phi(x, r) = f(x) + r \cdot P(x)$$

where:

- $r > 0$ is the penalty parameter (large),
- $P(x)$ is a penalty function like:
 - $\sum c_i(x)^2$ for equality constraints,
 - $\sum \max(0, g_i(x))^2$ for inequality constraints.
- Types:
 - Quadratic Penalty: $P(x) = \sum c_i(x)^2$
 - Exterior Penalty: Solution is driven toward feasibility from outside the feasible region.
 - Interior Penalty (used when combined with barrier): Not typically standalone.
- Properties:
 - As $r \rightarrow \infty$, the minimizer of $\Phi(x, r)$ approaches the solution of the constrained problem.
 - May cause ill-conditioning due to large r , making the numerical solution unstable.

If you go outside you have to pay a fine.

$r \rightarrow \infty$

Solve

r such that $\frac{1}{r}$ is negligible

Barrier Methods

- Core Idea: Prevent the optimization trajectory from leaving the feasible region by making it *infinitely costly* to approach the boundary of the constraint set.
- Formulation: For inequality-constrained problem:

$$\min f(x) \quad \text{subject to } g_i(x) \leq 0$$

The barrier function approach solves:

$$\min \Phi(x, \mu) = f(x) + \mu \cdot B(x)$$

where:

- $\mu > 0$ is a small parameter (shrinking to 0),
- $B(x) = -\sum \log(-g_i(x))$ for inequality constraints $g_i(x) < 0$
- Characteristics:
 - Operates *within* the feasible region (interior-point method).
 - As $\mu \rightarrow 0$, the barrier term vanishes, and the solution approaches the constrained minimum.
 - The trajectory stays strictly feasible, i.e., doesn't touch constraint boundaries.

Comparison Summary

Penalty Method	Barrier Method
Equality & inequality	Inequality
Penalizes violations	Penalizes proximity to constraint boundary
Always feasible during optimization	Always feasible (interior of feasible region)
$r \rightarrow \infty$	$\mu \rightarrow 0$
Prone to instability (ill-conditioning)	More stable trajectory but complex formulation

Use Cases

- Penalty methods are often used in early iterations or when constraint violations are tolerable.

- Barrier methods are fundamental in interior-point algorithms used in nonlinear and convex programming.

These methods are frequently used to construct *augmented Lagrangians* and interior-point algorithms and serve as foundations for many modern solvers.

- ① $\min x^2, \quad x \geq 2$
- ② $\min x^2 + y^2, \quad x + y \geq 1$
- ③ $\min x + y, \quad x^2 + y^2 \leq 1$



- ① $\min x^2, \quad x < 2$
- ② $\min x^2 + y^2, \quad x, y > 0$
- ③ $\min x + y, \quad x, y > 0$
 $x + y < 1$

ODS by Manvendra

Frank-Wolfe Algorithm (also known as Conditional Gradient Method)

The Frank-Wolfe algorithm is an iterative first-order optimization technique for constrained convex optimization problems. It is particularly suited for problems where projection onto the feasible set is computationally expensive, but linear optimization over it is efficient.

Problem Setup

Minimize a smooth convex function $f(x)$ over a compact convex set $\mathcal{D} \subset \mathbb{R}^n$:

$$\min_{x \in \mathcal{D}} f(x)$$

Core Idea

Replace the difficult constraint with a linear approximation at each step.

Algorithm Steps

1. Initialization: Choose $x^{(0)} \in \mathcal{D}$

2. Iterative step (for $k = 0, 1, 2, \dots$):

- Compute the gradient: $\nabla f(x^{(k)})$
- Solve a linear subproblem:

$$s^{(k)} = \arg \min_{s \in \mathcal{D}} \nabla f(x^{(k)})^\top s$$

- Update the current point:

$$x^{(k+1)} = x^{(k)} + \gamma_k (s^{(k)} - x^{(k)})$$

where $\gamma_k \in [0, 1]$ is the step size. Options:

- Fixed step size: $\gamma_k = \frac{2}{k+2}$
- Line search: minimize $f(x^{(k)} + \gamma(s^{(k)} - x^{(k)}))$

Key Characteristics

- Avoids projections: Uses linear minimization instead of projection.
 - Descent method: Each step decreases the objective function.
 - Convergence: Sublinear rate $\mathcal{O}(1/k)$ for convex problems.
-

Use Cases

- Large-scale convex problems
 - Problems with complicated feasible sets where projection is costly (e.g., Lasso, matrix completion, transportation problems)
-

Advantages

- Simplicity
 - Efficient per iteration (only requires gradient and linear optimization)
 - Natural sparsity in solutions (since each $s^{(k)}$ is often an extreme point of $\mathcal{D}$)
-

Limitations

- Slow convergence compared to second-order methods
 - May require many iterations for high accuracy
-

Application Example (From Fractal 2 Problem 6)

Objective:

$$\min_{x,y \geq 0} x^3 + y^3 \quad \text{s.t.} \quad 3x + y \leq 3, \quad 5x - 3y \leq 5$$

Frank-Wolfe steps:

1. Linearize $f(x, y) = x^3 + y^3$ at $(x^{(k)}, y^{(k)})$
2. Solve LP over feasible region
3. Move towards solution of LP using optimal step size

Performed for two iterations in assignment.

Let me know if you want convergence proofs or code snippet implementations for this.

1

$$\textcircled{1} \min x^2 + y^2$$

$$x + y \leq 1, \quad x, y \geq 0$$

2

$$(x-1)^2 + (y-2)^2$$

$$x + y = 1, \quad x, y \geq 0$$

End of ODS

X